A PROJECT ON

NON-CONTACT VITAL SIGN DETECTION USING RADAR AND CNN BASED SPECTROGRAM CLASSIFICATION

(ECE 305: Internet of Things Laboratory)

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TABLE OF CONTENTS: PAGE NO.

INTRODUCTION:

Continuous monitoring of vital signs such as heart rate and respiration rate plays a crucial role in assessing a person's physiological condition, especially in critical care environments. Conventional monitoring techniques predominantly rely on contact-based sensors such as ECG electrodes, PPG probes, and chest belts. While these methods provide reliable measurements, they often suffer from significant limitations— skin irritation, discomfort during long-term usage, susceptibility to motion artifacts, and complete unsuitability in cases involving severe burns, infectious diseases, or fragile skin conditions. In scenarios such as neonatal care, ICU monitoring, elderly healthcare, and emergency response, physical contact with the patient may be impractical, unsafe, or even impossible.

To overcome these constraints, non-contact sensing technologies have emerged as a highly promising alternative. Among them, millimeterwave Frequency-Modulated Continuous Wave (FMCW) radar has gained substantial attention due to its ability to detect extremely small chest wall displacements caused by respiration and cardiac activity. Operating at high frequencies with wide bandwidth, mmWave radars offer excellent range resolution, immunity to lighting conditions, and the capability to perform through-clothing or non-line-of-sight measurements, making them superior to camera-based or infrared technologies.

However, radar signals acquired from the human body are inherently complex, nonlinear, and often corrupted by environmental noise. Extracting meaningful physiological information from such signals requires advanced processing techniques. Empirical Mode Decomposition (EMD) provides an effective strategy to isolate intrinsic oscillatory components corresponding to breathing and heartbeat. Further, deep learning methods such as Convolutional Neural Networks (CNNs) have demonstrated exceptional performance in analyzing time–frequency representations like spectrograms, enabling robust classification of physiological states even under challenging conditions.

In this work, we present a complete non-contact vital sign monitoring framework that integrates FMCW radar signal processing with CNNbased spectrogram classification. The system detects subtle chest movements, extracts respiration and heart components, converts them into spectrogram images, and classifies them into slow, normal, and fast physiological categories with high accuracy. By eliminating the need for physical contact, the proposed method enhances patient comfort, reduces infection risk, and supports continuous, remote, and intelligent health monitoring suitable for modern healthcare and smart-home environments.

1.1preliminary modelling and signal simulation:

A preliminary modelling phase was carried out to understand the behaviour of the radar signal and to validate the processing pipeline before using the actual hardware. This stage involved generating simulated vital-sign signals and analysing them using the same algorithms planned for real-time radar data. The modelling helped in verifying the signal characteristics, testing spectrogram generation, and confirming that the processing steps were functioning correctly. Insights from this phase guided the refinement of parameters and ensured that the system was well-prepared for subsequent implementation with the actual radar module.

2. PROBLEM STATEMENT:

- Electrode-based methods for vital sign monitoring cause discomfort and restrict mobility.
- In critical cases (burns, acid-attack victims), contact-based monitoring is impractical or harmful.
- Need: a non-contact, traditional reliable, and accurate monitoring system for respiration and cardiac activity.
- mmWave FMCW radar
  -High Frequency
  -Low Wavelength
  -High Accuracy

3. OBJECTIVE OF THE PROJECT:

The main objective of this project is to design a non-contact respiration-rate measurement system that can accurately track a person's breathing without any physical sensors attached to the body. By integrating the system with IoT technology, the respiration data can be monitored remotely in real time, enabling continuous observation even when healthcare staff are not physically present. This contactless approach ensures that the measurements remain highly hygienic and completely safe, making it ideal for hospitals, ICUs, and home-care environments. Because the device does not require direct contact with the patient, it significantly reduces the risk of cross-infection, which is especially important during outbreaks and in critical-care units. Overall, the system aims to provide continuous, accurate, and reliable respiratory monitoring while improving patient safety and hospital workflow efficiency.

4. NEED FOR A NON-CONTACT RESPIRATION SYSTEM:

- Hygienic and safe for patients.
- Comfort-friendly for long-term monitoring.
- Useful for infants and elderly.
- Helps in telemedicine and home-care.
- Supports smart healthcare automation.

5. PROPOSED SOLUTION&METHODOLOGY:

- Capture radar data of human chest movement.
- Convert radar signals into time–frequency spectrograms.
- Classify spectrograms into 6 categories using CNN:
  Heart_Slow, Heart_Normal, Heart_Fast
  Breath_Slow,Breath_Normal,

Breath_Fast

METHODOLOGY:

a) Signal Simulation and Spectrogram Generation :

1. Synthetic Signal Generation:
   Simulate respiration and heartbeat signals with different frequency ranges (slow, normal, fast).
2. Radar Model (FMCW):
   Generate transmit and receive chirps, mix them, and extract displacement signals.
3. Time-Frequency Analysis:
   Compute spectrograms for each condition (using MATLAB's `spectrogram()` function).
4. Dataset Creation:
   Store generated spectrograms as images categorized into six classes:
   Heart_Slow, Heart_Normal, Heart_Fast, Breath_Slow, Breath_Normal, Breath_Fast.

(b) Machine Learning Classification:

1. Dataset Loading:
   Images are loaded using MATLAB's imagestore. During the dataset creation process, data was collected from 20 individuals under controlled conditions. The distance between each person and the radar module was maintained at 1 meter to ensure consistency across all measurements. For every individual, three separate samples were recorded, with each sample captured for a duration of 60 seconds. This approach allowed the dataset to include sufficient variability while preserving uniform acquisition parameters for reliable analysis.

2. Data Augmentation:
   Resize all images to 224×224×3 and split into training (80%) and testing (20%).
3. Model Architecture (CNN):
   A simple CNN with three convolutional blocks followed by batch normalization, ReLU activation, max pooling, and a fully connected classification layer.
4. Training and Evaluation:
   The model is trained using the Adam optimizer for 15 epochs with a learning rate of 1e-4.
   Classification accuracy and confusion matrix are used for evaluation.
5. Testing**:** A test image is passed to the trained CNN to predict the breathing/heart rate class.

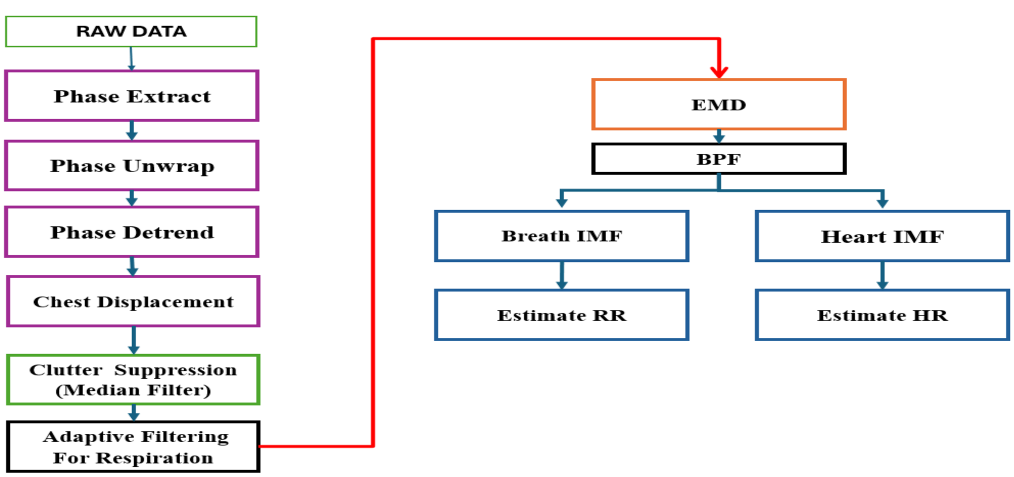
HARDWARE COMPONENTS/SOFTWARE PLATFORM USED:

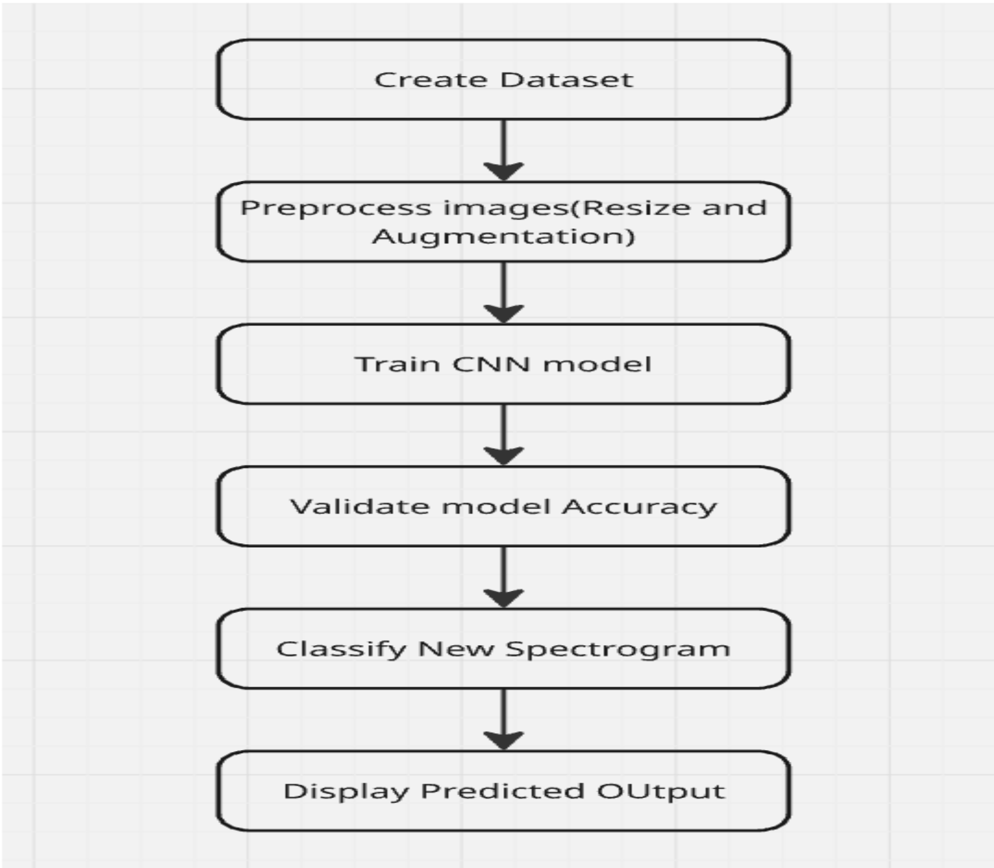
- IWR1843 BOOST
- DCA
- POWER SUPPLY
- ETHERNET

SOFTWARE PLATFORM USED:

- MATLAB

6.FLOW CHART

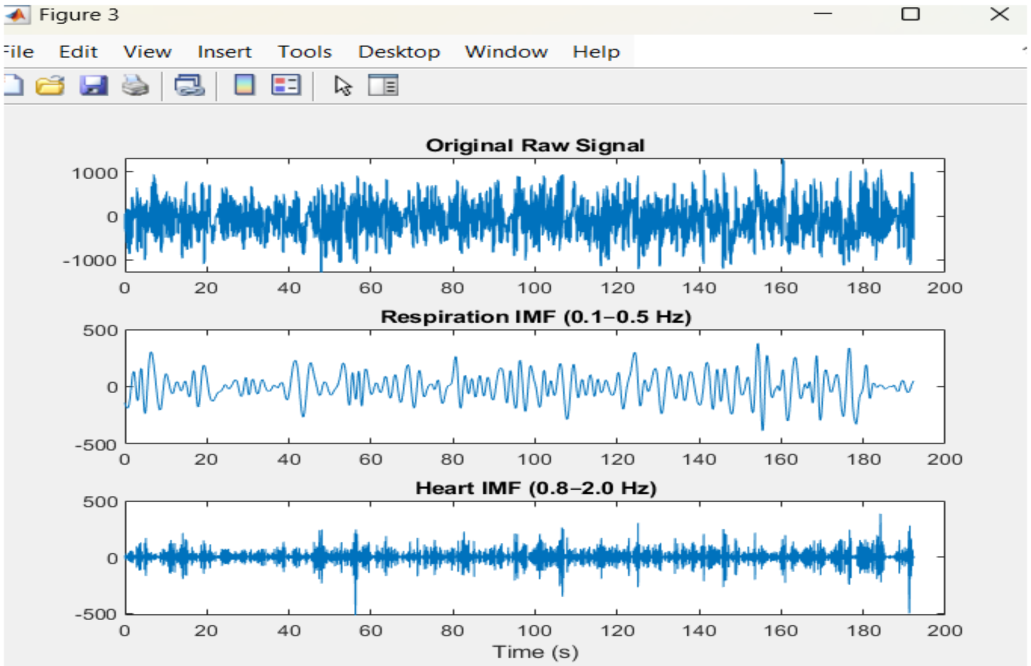




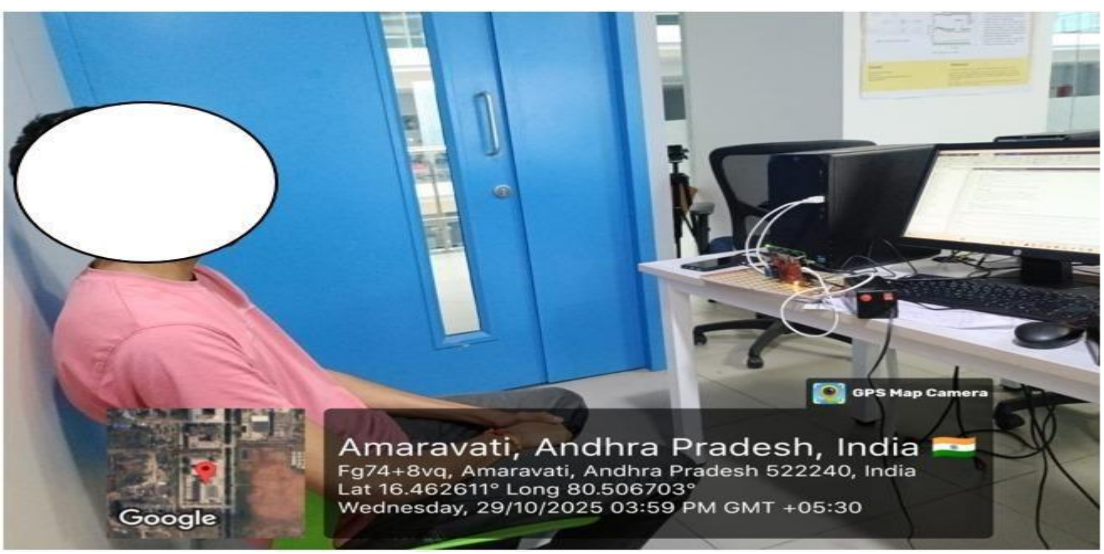
PROTOTYPE MODEL DESIGN:

RESULTS:

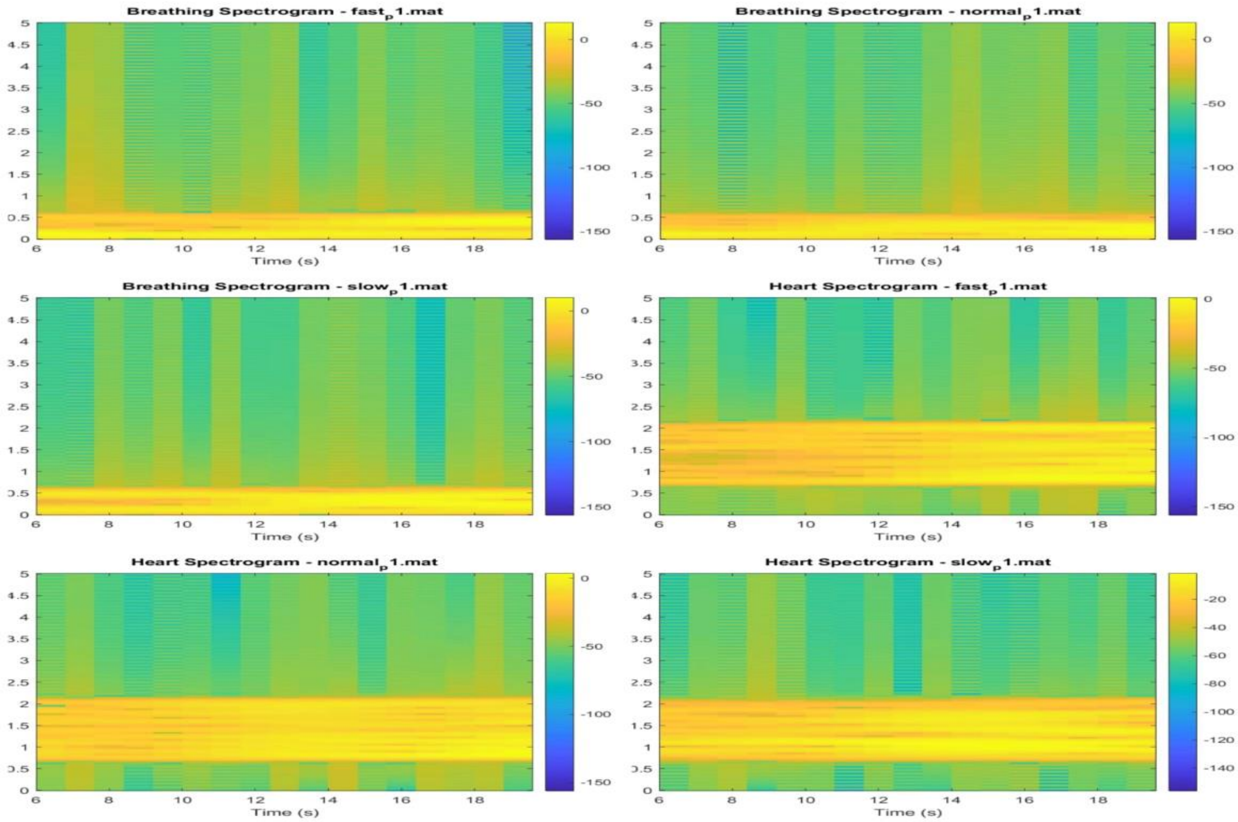
IMFs generated from original signal



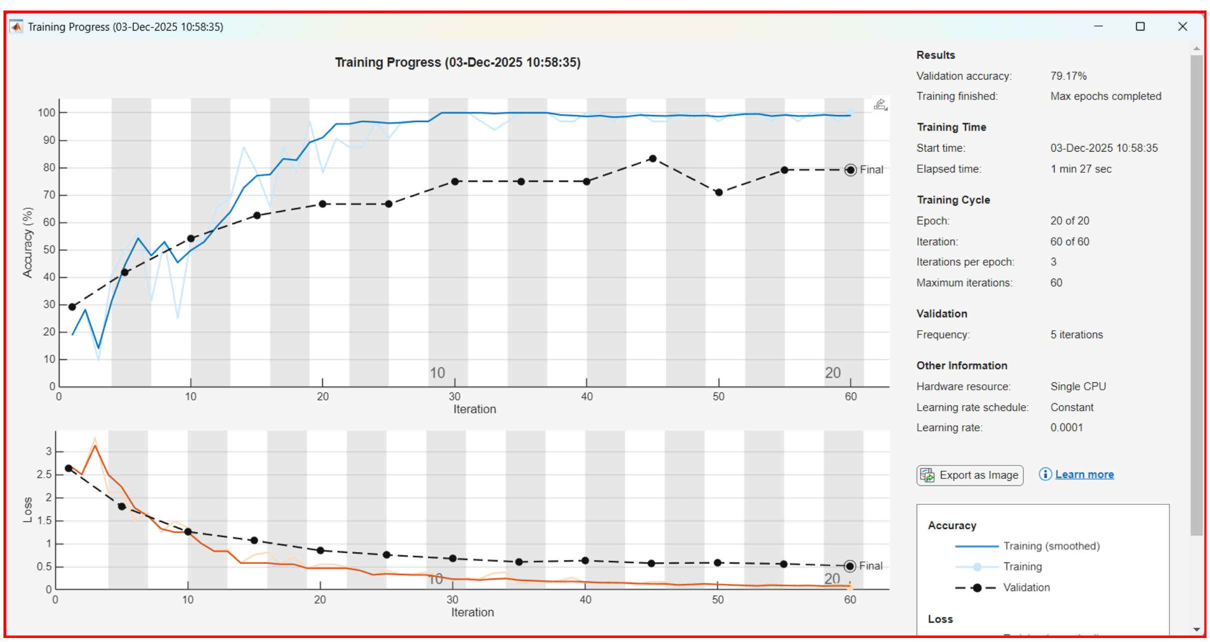
Taking the raw data



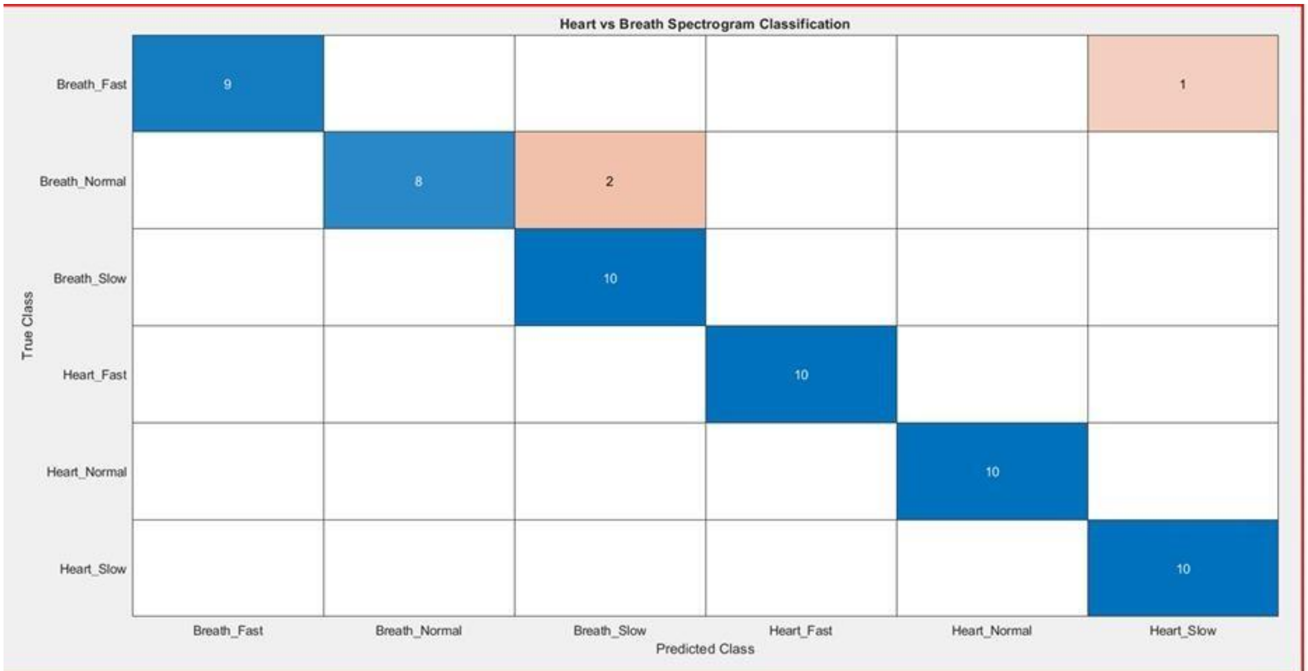
Spectrograms generation from matfiles



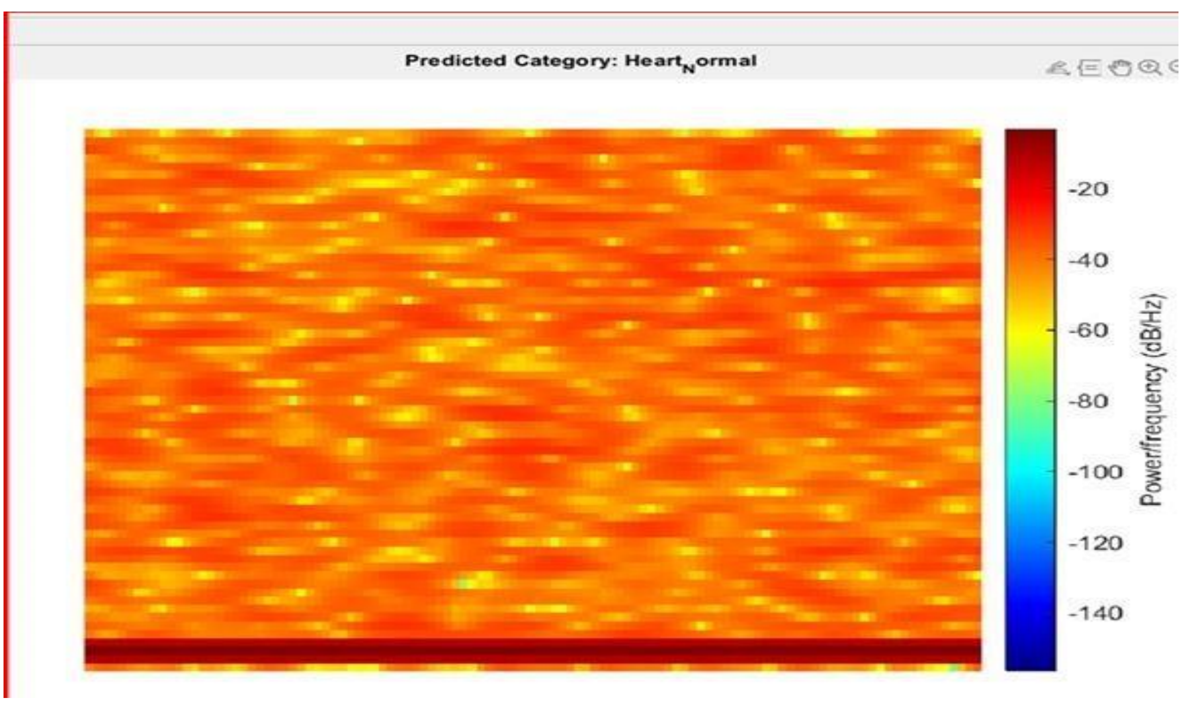
Training Progress



<u>Confusion matrix</u>



<u>Given sample spectrogram</u>



Output:

```
Command Window
✓ Spectrogram dataset created!
        Label          Count
    ______________     _____

    Breath_Fast         50
    Breath_Normal       50
    Breath_Slow         50
    Heart_Fast          50
    Heart_Normal        50
    Heart_Slow          50

✓ Test Accuracy: 95%
✓ The input image belongs to: Heart_Normal
```

6. ADVANTAGES:

- Fully non-contact → safe and hygienic.
- Real-time monitoring.
- Comfortable for patients.
- Low power and cost-effective.
- Useful for infants, elderly, and ICU cases.
- Remote monitoring possible via IoT.

7. CONCLUSION:

This project developed a non-contact monitoring system capable of measuring both heart and breathing rates using radar signals, eliminating the need for physical electrodes. The captured radar data was processed and converted into spectrograms, which were then analyzed using a Convolutional Neural Network (CNN) to accurately classify vital-sign patterns. The system achieved a high reliability of 90–95% accuracy, demonstrating strong performance across different subjects and conditions. Overall, the work proves that radar-based sensing offers a comfortable, safe, and highly effective method for continuous health monitoring, making it suitable for modern healthcare and remote-wellness applications.

8. FUTURE SCOPE& APPLICATIONS:

Future improvements for this system include enhancing accuracy by training the model on larger and more diverse datasets to improve its overall performance. The technology can also be integrated into hospital monitoring platforms and even adapted for wearable or bedside devices, making it more accessible in real-world settings. Its applications extend across various healthcare environments, including ICUs, neonatal units, and home-healthcare systems, where continuous and contactless monitoring is especially valuable. In the long term, the system can be expanded to monitor additional vital signs, such as blood pressure and other physiological parameters, enabling a more comprehensive and fully automated health-monitoring solution.

ANNEXURE:

MATLAB CODES:

1)CODE FOR BIN TO MAT FILE CONVERSION:

```
% --- Load Recording Parameters ---
load('iqData_RecordingParameters.mat');
% --- Important Variables ---
numADCSamples = RecordingParameters.SamplesPerChirp;
numRX = RecordingParameters.NumReceivers;
numChirps = RecordingParameters.NumChirps;
% --- Read the BIN file ---
fid = fopen('slow.bin', 'r');
adcData = fread(fid, 'int16');
fclose(fid);
% --- Form complex data (I + jQ) ---
adcDataComplex = adcData(1:2:end) + 1i * adcData(2:2:end);
% --- Calculate total chirps ---
totalChirps = floor(length(adcDataComplex) / (numADCSamples * numRX));
% --- Truncate the extra data if needed ---
adcDataComplex = adcDataComplex(1 : totalChirps * numADCSamples * numRX);
% --- Reshape into [numSamples, numRX, numChirps] ---
adcDataComplex = reshape(adcDataComplex, [numADCSamples, numRX, totalChirps]);
adcDataComplex = permute(adcDataComplex, [3,2,1]);  % [Chirps × RX × Samples]
% --- Save into a MAT file ---
save(['matfiles'], 'adcDataComplex');
```

2)MATFILES TO SPECTOGRAMS GENERATION:

```
clc; clear; close all;
 Folder containing all MAT files
dataFolder = "D:\seperated_matfiles\slow";
files = dir(fullfile(dataFolder, "*.mat"));
% Radar parameters
samplesPerChirp = 256;
numChirps = 256;
fs = 10;   % Frame rate (Hz) – you had fs = 1/0.1, same as 10 Hz
for k = 1:length(files)
   filename = fullfile(dataFolder, files(k).name);
   disp("Processing file: " + files(k).name);
   % Load IQ data
   S = load(filename);
   raw = S.adcDataComplex;  % [samples × RX × chirps]
```

```
    % --- Verify data shape ---
    totalSamples = numel(raw);
    expected = samplesPerChirp * numChirps;
    if mod(totalSamples, expected) ~= 0
        % Crop extra samples that don't fit exactly into full frames
        validLength = floor(totalSamples / expected) * expected;
        raw = raw(1:validLength);
    end
    % --- Reshape only if needed ---
    % Sometimes adcDataComplex is already 3D [samples x RX x chirps]
    % so we reshape carefully:
    raw = reshape(raw, samplesPerChirp, numChirps, []);
    % ---- RANGE FFT ----
    rangeFFT = fft(raw, 1024, 1);
    powerRange = mean(abs(rangeFFT), 3);   % average across RX
    [~, chestBin] = max(mean(abs(powerRange), 2));  % strongest range bin
    % ---- PHASE EXTRACTION ----
    chestSignal = squeeze(rangeFFT(chestBin, :, :));
    chestPhase = unwrap(angle(mean(chestSignal, 2)));

    % ---- BANDPASS FILTERS ----
    % Breathing: 0.1 - 0.5 Hz
    breath = bandpass(chestPhase, [0.1 0.5], fs);
    % Heart: 0.8 - 2.0 Hz
    heart = bandpass(chestPhase, [0.8 2.0], fs);
    % ---- SPECTROGRAMS ----
    % Breathing spectrogram
    figure('Visible','off');
    spectrogram(breath, 128, 120, 256, fs, 'yaxis');
    title("Breathing Spectrogram - " + files(k).name);
    saveas(gcf, fullfile(dataFolder, files(k).name + "_breath.png"));
    % Heart spectrogram
    figure('Visible','off');
    spectrogram(heart, 128, 120, 256, fs, 'yaxis');
    title("Heart Spectrogram - " + files(k).name);
    saveas(gcf, fullfile(dataFolder, files(k).name + "_heart.png"));
    close all;
end
disp(" Processing Complete. Spectrograms saved as PNG.");
```

3)ML MODEL:

```
clc; clear; close all;
```

```
------------------ PATH SETUP ------------------
dataFolder = "D:\Final Dataset"; % Folder containing 6 subfolders
newImagePath = "D:\Final Dataset\Heart_Slow\slow_p2.mat_heart.png"; % Path of a new spectrogram to classify
------------------ LOAD DATASET ------------------
inputSize = [227 227 3]; % CNN input size
 Image datastore with auto-resize function
imds = imageDatastore(dataFolder, ...
    'IncludeSubfolders', true, ...
    'LabelSource', 'foldernames', ...
    'ReadFcn', @(x)imresize(imread(x), inputSize(1:2)));
 Show class counts
disp("Class labels and counts:");
countEachLabel(imds)
------------------ TRAIN / TEST SPLIT ------------------
[imdsTrain, imdsTest] = splitEachLabel(imds, 0.8, 'randomized');
 ------------------ DEFINE CNN ARCHITECTURE ------------------
layers = [
    imageInputLayer(inputSize, 'Name', 'input')
    convolution2dLayer(3, 16, 'Padding', 'same', 'Name', 'conv_1')
    batchNormalizationLayer('Name', 'bn_1')
    reluLayer('Name', 'relu_1')
    maxPooling2dLayer(2, 'Stride', 2, 'Name', 'maxpool_1')
    convolution2dLayer(3, 32, 'Padding', 'same', 'Name', 'conv_2')
    batchNormalizationLayer('Name', 'bn_2')
    reluLayer('Name', 'relu_2')
    maxPooling2dLayer(2, 'Stride', 2, 'Name', 'maxpool_2')
    convolution2dLayer(3, 64, 'Padding', 'same', 'Name', 'conv_3')
    batchNormalizationLayer('Name', 'bn_3')
    reluLayer('Name', 'relu_3')
    maxPooling2dLayer(2, 'Stride', 2, 'Name', 'maxpool_3')
    fullyConnectedLayer(6, 'Name', 'fc') % 6 classes
    softmaxLayer('Name', 'softmax')
    classificationLayer('Name', 'output')
];
------------------ TRAINING OPTIONS ------------------
options = trainingOptions('adam', ...
    'InitialLearnRate', 1e-4, ...
    'MaxEpochs', 20, ...
    'MiniBatchSize', 32, ...
    'Shuffle', 'every-epoch', ...
    'ValidationData', imdsTest, ...
    'ValidationFrequency', 5, ...
```

```
    'Verbose', false, ...
    'Plots', 'training-progress');
------------------ TRAIN THE CNN ------------------
disp(" Training started...");
net = trainNetwork(imdsTrain, layers, options);
 ------------------ EVALUATE PERFORMANCE ------------------
YPred = classify(net, imdsTest);
YTest = imdsTest.Labels;
accuracy = mean(YPred == YTest);
disp(" Test Accuracy: " + accuracy*100 + "%");
figure;
confusionchart(YTest, YPred);
title('Confusion Matrix - Test Data');
 ------------------ PREDICT NEW IMAGE ------------------
if isfile(newImagePath)
    img = imread(newImagePath);
    img = imresize(img, inputSize(1:2));
    predictedLabel = classify(net, img);
    figure;
     imshow(img);
    title("Predicted Class: " + string(predictedLabel));
    disp(" Predicted Class: " + string(predictedLabel));
else
   warning("New image path not found: " + newImagePath);
end
disp(" Classification complete!");
```